Lecture 1

Groups, examples, and the landscape of finite groups

Objectives. *We abstract the group axioms from the behaviour of symmetry operations, assemble a working zoo of finite groups — cyclic, dihedral and symmetric — together with a first glimpse of the matrix groups of Part II, make precise when two groups are "the same", and prove the fundamental counting theorem of the subject, Lagrange's theorem.*

1.1 From symmetry operations to the group axioms

Symmetry, in physics as in geometry, is always symmetry *under something*: a transformation that leaves the relevant structure unchanged. The structure may be a geometric figure, a crystal lattice, a molecule's equilibrium configuration, or — as we shall see at the end of Part I — the Hamiltonian of a physical system. The first task of this course is to extract, from the behaviour of such transformations, a small set of axioms, and then to develop the mathematics of those axioms far enough that it starts returning concrete physical predictions: degeneracies of energy levels, patterns of vibrational spectra, selection rules.

We begin with the simplest nontrivial example. Let Δ be an equilateral triangle in the plane, its vertices labelled $1, 2, 3$. A *symmetry operation* of Δ is a distance-preserving map of the plane that carries Δ onto itself. There are exactly six (Figure 1.1):

- the identity map e, which moves nothing;
- the rotations r and r^2 about the centroid, through $2\pi/3$ and $4\pi/3$ anticlockwise;
- the reflections s_1, s_2, s_3 in the three axes joining a vertex to the midpoint of the opposite side.

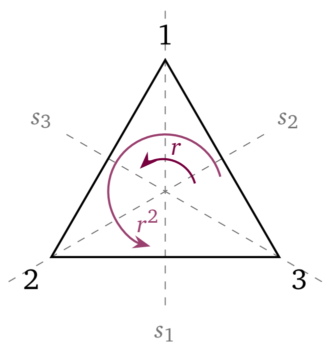


Figure 1.1: The six symmetry operations of the equilateral triangle: the identity, two rotations about the centroid, and reflections in the three dashed axes. The axis s_i passes through vertex i.

Each operation is determined by what it does to the vertices. Writing $g\colon i \mapsto j$ when g carries vertex i to the position of vertex j. For instance,

$$r\colon 1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 1, \qquad s_1\colon 1 \mapsto 1,\ 2 \mapsto 3,\ 3 \mapsto 2,$$

and similarly for the rest.

Two operations performed in succession are again a symmetry operation. Throughout the course we write composition *right to left*: gh means "first h, then g". So $(gh)(x) = g\big(h(x)\big)$ for every point x.

Trace the vertices:

$$rs_1\colon 1 \xmapsto{s_1} 1 \xmapsto{r} 2, \quad 2 \mapsto 3 \mapsto 1, \quad 3 \mapsto 2 \mapsto 3,$$

So rs_1 fixes vertex 3 and swaps $1, 2$, which is s_3. The same computation in the other order gives $s_1 r = s_2$. Hence

$$rs_1 = s_3 \neq s_2 = s_1 r.$$

The order in which symmetry operations are performed *matters*. This noncommutativity is not a defect. It is the source of most of the structure we study in this course.

Four features of this example are general. The composition of two symmetry operations is a symmetry operation. Composition of operations is associative. The identity map is a symmetry operation. Every symmetry operation can be "undone" by an *inverse* operation. That is all we need in order to define symmetry abstractly.

1.2 The definition of a group

Definition 1.1 (Group)**.** A *group* is a set G together with a map $G \times G \to G$, $(g, h) \mapsto gh$, called the *group multiplication*, such that:

(G1) *Associativity:* $(gh)k = g(hk)$ for all $g, h, k \in G$.

(G2) *Identity:* there exists $e \in G$ with $eg = ge = g$ for all $g \in G$.

(G3) *Inverses:* for every $g \in G$ there exists $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

Remark 1.2 (Conventions)**.** The property that $gh \in G$ for all $g, h \in G$ is called *closure*. We write the multiplication by juxtaposition and call gh the product, even when the operation is composition of maps or addition of numbers. We denote the identity element e (German *einselement*); the physics and chemistry literature often writes E, and algebra 1. By associativity, products such as ghk need no brackets. Powers $g^n = g \cdots g$ (n factors), $g^0 = e$, $g^{-n} = (g^{-1})^n$ obey the usual rules of exponents.

Definition 1.3 (Abelian group)**.** A group G is *abelian* (or commutative) if $gh = hg$ for all $g, h \in G$. For abelian groups one sometimes uses additive notation: $g + h$ for the product, 0 for the identity, and $-g$ for the inverse.

Example 1.4. (i) The integers $\mathbb{Z}$ together with addition form an abelian group: 0 is the identity, and the inverse of n is $-n$. The same holds for the real and complex numbers, $\mathbb{R}$ and $\mathbb{C}$, together with addition. (ii) The nonzero rationals $\mathbb{Q} \setminus \{0\}$ together with multiplication form an abelian group, as do $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$; the two-element set $\{1, -1\}$ together with multiplication is a group too. (iii) The natural numbers $\mathbb{N}$ together with addition are *not* a group (no inverses), and $\mathbb{Z}$ together with multiplication is not a group (only 1 and -1 have inverses). (iv) The six symmetry operations of the triangle together with composition form a *nonabelian* group; we showed this above.

Proposition 1.5 (Elementary consequences of the axioms). *Let G be a group. Then:*

(i) *the identity element is unique;*
(ii) *each $g \in G$ has a unique inverse;*
(iii) *(cancellation) $gh = gk$ implies $h = k$, and $hg = kg$ implies $h = k$;*
(iv) *$(gh)^{-1} = h^{-1}g^{-1}$ and $(g^{-1})^{-1} = g$.*

Proof. (i) If e and e' both satisfy the identity axiom, then $e = ee' = e'$. (ii) If h and h' are both inverse to g, then $h = he = h(gh') = (hg)h' = eh' = h'$. (iii) Multiply $gh = gk$ on the left by g^{-1} and use associativity: $h = eh = (g^{-1}g)h = g^{-1}(gh) = g^{-1}(gk) = k$; right cancellation is analogous. (iv) $(h^{-1}g^{-1})(gh) = h^{-1}(g^{-1}g)h = h^{-1}h = e$, and similarly $(gh)(h^{-1}g^{-1}) = e$, so $h^{-1}g^{-1}$ is *an* inverse of gh and hence, by the uniqueness just proved, *the* inverse. Finally g inverts g^{-1}, so $(g^{-1})^{-1} = g$. ∎

Note the order reversal $(gh)^{-1} = h^{-1}g^{-1}$: to undo "socks, then shoes" one has to remove the shoes first. A composite transformation is undone by performing its steps in *reverse order*.

Definition 1.6 (Order). The *order* of a group G, written $|G|$, is the number of its elements; G is *finite* if $|G| < \infty$. The *order* of an element $g \in G$, written $\operatorname{ord}(g)$, is the smallest integer $n \geq 1$ with $g^n = e$, if such an n exists; otherwise g has infinite order.

Proposition 1.7. *Let G be any group and let $g \in G$ have finite order $n = \operatorname{ord}(g)$. Then for $m \in \mathbb{Z}$,*

$$g^m = e \iff n \mid m,$$

and the powers $e, g, g^2, \dots, g^{n-1}$ are distinct. If G is finite, then every element has finite order.

Proof. If $n \mid m$, say $m = qn$, then $g^m = (g^n)^q = e$. Conversely, if $g^m = e$, write $m = qn + s$ with $0 \leq s < n$ (division with remainder); then $e = g^m = (g^n)^q g^s = g^s$, and minimality of n forces $s = 0$. Distinctness follows: if $g^a = g^b$ with $0 \leq b < a < n$ then $g^{a-b} = e$ with $0 < a - b < n$, contradicting minimality.

For the last statement, let G be finite. The elements $g, g^2, g^3, \dots$ cannot all be distinct in a finite set, so $g^a = g^b$ for some $a > b \geq 1$; cancellation gives $g^{a-b} = e$ with $a - b \geq 1$, so g has finite order. ∎

In the triangle group, the rotation r has order 3 and each reflection s_i has order 2: a reflection is its own inverse.

1.3 Examples

In this section we meet three families of finite groups: cyclic, dihedral and symmetric groups. These groups will be our laboratory for the first part of the course.

1.3.1 Cyclic groups

Definition 1.8 (Cyclic group). A group G is *cyclic* if there is an element $g \in G$, called a *generator*, such that every element of G is a power of g:

$$G = \{\, g^k : k \in \mathbb{Z} \,\}.$$

A cyclic group having n elements is denoted C_n.

If a generator g has order n, then by Proposition 1.7 the powers $e, g, g^2, \dots, g^{n-1}$ are distinct and exhaust the group, and $g^a g^b = g^{a+b}$ with the power read modulo n. Concrete examples of the abstract group C_n:

(a) the rotations of the plane through multiples of $2\pi/n$ about a chosen point, under composition;
(b) the n-th roots of unity $\{1, \omega, \omega^2, \dots, \omega^{n-1}\} \subset \mathbb{C}$, where $\omega = e^{2\pi i/n}$, under multiplication;
(c) the residues $\{0, 1, \dots, n-1\}$ under addition modulo n, written $\mathbb{Z}_n$.

That these examples are "the same group" is made precise in Section 1.3.4. Every cyclic group is abelian, since $g^a g^b = g^{a+b} = g^b g^a$. The integers $\mathbb{Z}$ under addition are the *infinite* cyclic group: every element is a multiple of the generator 1.

1.3.2 Dihedral groups

Definition 1.9 (Dihedral group)**.** For $n \geq 3$, the *dihedral group* D_n is the group of symmetry operations of a regular n-gon, that is, of all distance-preserving maps of the plane carrying the n-gon onto itself, under composition.

Some books write D_{2n} for this group, where $2n$ gives its order rather than the number of corners of the polygon. We shall always label by the number of corners.

Proposition 1.10 (Structure of D_n)**.** *Let r denote the rotation through $2\pi/n$ about the centre of the n-gon, and s — the reflection in a chosen symmetry axis. Then $|D_n| = 2n$, and*

$$D_n = \{\, e, r, r^2, \dots, r^{n-1},\ s, rs, r^2 s, \dots, r^{n-1} s \,\},$$

where the first n elements are the rotations and the remaining n are the reflections. The generators obey

$$r^n = s^2 = e, \qquad srs^{-1} = r^{-1} = r^{n-1}, \tag{1.1}$$

and consequently $sr^k = r^{-k}s = r^{n-k}s$ for all k and

$$(r^a s^\varepsilon)(r^b s^\delta) = r^{\,a+(-1)^\varepsilon b}\, s^{\,\varepsilon+\delta}, \qquad a, b \in \mathbb{Z}, \quad \varepsilon, \delta \in \{0, 1\}, \tag{1.2}$$

where the power of r is read modulo n and that of s modulo 2.

Proof. Every symmetry operation of the polygon leaves its centre fixed, and is therefore orthogonal: a rotation or a reflection. The possible rotations are through 0, $2\pi/n$, $4\pi/n$, …, $2(n-1)\pi/n$ anticlockwise, and the possible reflections are in the axes shown in Figure 1.2. There are n rotations and n reflections in all, so $|D_n| = 2n$.

Next we show that every reflection can be written as $r^k s$ with $0 \leq k < n$. If t is any reflection, then ts preserves orientation, so it is a rotation: $ts = r^k$ for some k. Using $s^2 = e$ we get $t = r^k s$. This also shows that the listed $2n$ elements exhaust D_n.

The conjugation relation remains. Compute how srs^{-1} acts on the plane: $s^{-1} = s$ reflects, r rotates by $2\pi/n$ anticlockwise, and s reflects back. A rotation viewed in a mirror is a rotation through the same angle in the opposite sense, so $srs^{-1} = r^{-1} = r^{n-1}$. (One can also check this by tracing two adjacent vertices, or by multiplying 2×2 orthogonal matrices.) Iterating gives $sr^k s^{-1} = r^{-k}$, i.e. $sr^k = r^{-k}s$, where $r^{-k} = r^{n-k}$. Finally (1.2) follows by moving all s's to the right: when $\varepsilon = 1$, pushing s past r^b flips the sign of b. ∎

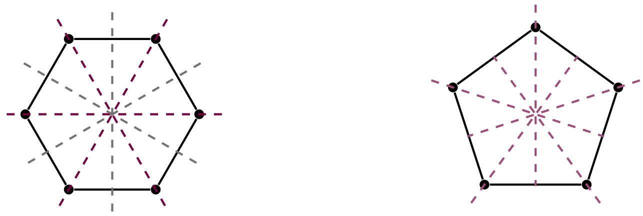


Figure 1.2: Reflection axes. When n is even the axes come in two kinds: through two opposite vertices (coloured) and through two opposite edge midpoints (grey), $n/2$ of each. When n is odd all n axes are of one kind: each runs through a vertex and the midpoint of the opposite edge. In both cases there are exactly n axes.

The rule (1.2) makes computation in D_n entirely mechanical. For $n = 3$, the group D_3 is precisely the six-element triangle group of Section 1.1. Choosing $s = s_1$ gives $s_2 = r^2 s$ and $s_3 = rs$. This agrees with the identities $rs_1 = s_3$ and $s_1 r = s_2$ derived there.

Tutorial

Example 1.11 (Cayley table of D_3)**.** List the elements in the order e, r, r^2, s, rs, r^2s: the left factor g down the first column, the right factor h along the top row. The product gh then stands where row g meets column h. Rule (1.2) yields the *Cayley table* (multiplication table)

gh	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

Notice that each row and each column contains every element exactly once (a *Latin square*). This holds in any finite group and follows from the cancellation law: by Proposition 1.5 the map $h \mapsto gh$ is injective, and on a finite set injective means bijective. Notice also that the table is not symmetric about its diagonal — that is a property of a nonabelian group.

Remark 1.12 (Generators and relations)**.** Proposition 1.10 says that every element of D_n is a word in r and s, and that every product of such words can be reduced using only the relations (1.1). One summarises this by the *presentation*

$$D_n = \left\langle r, s \;\middle|\; r^n = s^2 = e,\ srs^{-1} = r^{-1} \right\rangle.$$

Presentations are a compact and convenient way of writing down a group. In this course we shall use them freely, because all our groups are concrete. The general theory of groups given by arbitrary presentations is far more subtle: even deciding whether two words denote the same element can be algorithmically undecidable. We shall not need it.

1.3.3 Symmetric groups

Definition 1.13 (Symmetric group)**.** The *symmetric group* S_n is the set of all permutations (bijections, or rearrangements) $\sigma\colon \{1,\dots,n\} \to \{1,\dots,n\}$, together with their composition.

Proposition 1.14. *S_n is a group, and $|S_n| = n!$.*

Proof. Composites and inverses of permutations are again permutations, their composition is associative, and the identity map is also a (trivial) permutation; so S_n is a group. We specify a permutation by choosing $\sigma(1)$ (n ways), then $\sigma(2)$ ($n-1$ remaining ways), and so on: $n!$ permutations in all. ∎

A permutation can be recorded in *two-line notation*, for instance

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 2 & 5 \end{pmatrix} \in S_5,$$

meaning $\sigma(1) = 3$, $\sigma(2) = 4$ and so on. This notation is intuitive but bulky. We shall use the more compact *cycle notation*.

The symbol $(a_1\, a_2\, \dots\, a_k)$ denotes the *k-cycle* sending $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixing every other point. By convention 1-cycles are not written. So, for example, $(1\ 2\ 3)$ and $(1\ 2\ 3)(4)(5)$ record the same permutation of S_5, while the identity keeps its own symbol e.

Multiplying cycles is explained in the example below. For now we note only that cycles built from disjoint sets of points (*disjoint cycles*) commute: for instance $(1\,2)(3\,4\,5) = (3\,4\,5)(1\,2)$. A 2-cycle $(a\,b)$, which merely swaps a and b, is called a *transposition*.

Proposition 1.15 (Disjoint cycle decomposition)**.** *Every permutation $\sigma \in S_n$ can be written as a product of pairwise disjoint cycles. This decomposition is unique up to the order of the factors.*

Proof ($\star$). Follow the point 1 under iteration, $1, \sigma(1), \sigma^2(1), \dots,$ until the first repetition, $\sigma^j(1) = \sigma^k(1)$ with $0 \le j < k$. Injectivity of the permutation forces that first repetition to be the starting point, $\sigma^k(1) = 1$. So we have built a cycle $\big(1\ \sigma(1)\ \cdots\ \sigma^{k-1}(1)\big)$.

Now repeat the same procedure from the smallest point not yet visited, until every point has been visited. The cycles so produced are disjoint by construction. Their product is σ, since each point is moved by exactly one of them, in the same way as by σ itself. The decomposition is unique up to the order of the factors, because the cycle containing a given point a is forced — it must read $\big(a\ \sigma(a)\ \sigma^2(a)\ \cdots\big)$. ∎

Example 1.16. This example teaches how cycles are multiplied. Cycles are multiplied by tracing one point at a time. Composition is read *right to left*, so a point is moved first by the right-hand factor and the result then by the left-hand one. A cycle does not mention the points it fixes, so a point absent from a factor passes through it unchanged.

Multiply $(1\,2)$ and $(2\,3)$ in S_3. Take each point through both factors:

$$1 \xmapsto{(2\,3)} 1 \xmapsto{(1\,2)} 2, \qquad 2 \xmapsto{(2\,3)} 3 \xmapsto{(1\,2)} 3, \qquad 3 \xmapsto{(2\,3)} 2 \xmapsto{(1\,2)} 1.$$

The product sends $1 \mapsto 2$, $2 \mapsto 3$ and $3 \mapsto 1$, which is the cycle $(1\,2\,3)$. Running the same computation with the factors swapped gives

$$(1\,2)(2\,3) = (1\,2\,3), \qquad (2\,3)(1\,2) = (1\,3\,2).$$

The results differ, so S_3 is nonabelian. The computation should look familiar: it is the triangle computation of Section 1.1 in a new notation. There a rotation and a reflection refused to commute; here the two reflections $(1\,2)$ and $(2\,3)$, composed in the two orders, give the two different rotations. We return to this in Example 1.23.

Proposition 1.17. *Every permutation is a product of transpositions.*

Proof. By Proposition 1.15 every permutation can be written as a product of disjoint cycles. It remains to check that each cycle can be written as a product of transpositions:

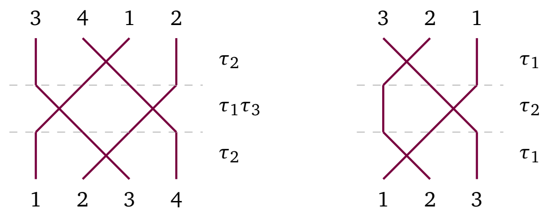
$$(a_1\, a_2\, \dots\, a_k) = (a_1\, a_k)(a_1\, a_{k-1}) \cdots (a_1\, a_2).$$

This follows from the method of multiplying cycles described in Example 1.16. ■

The decomposition into transpositions is not unique. For instance $e = (1\,2)(1\,2)$ uses two transpositions, while e itself uses none. The *parity* of the number of factors, however, does not depend on the decomposition. To see this, let us draw the permutation.

Picture a permutation $\sigma \in S_n$ as a braid of n strands (Figure 1.3), read from bottom to top. The strand starting at i ends at $\sigma(i)$. Two strands starting at $i < j$ cross exactly when $\sigma(i) > \sigma(j)$. Such a pair is called an *inversion*, and the number of inversions is written $\operatorname{inv}(\sigma)$.

Every braid can be threaded from *elementary* layers, each swapping just two neighbouring strands. Such a layer is an adjacent transposition $\tau_i = (i\ \ i{+}1)$, and multiplying on the left adds a new layer on top. That is how the next proof works: we pull the strands about, but never cut them.



$(1\,3)(2\,4) = \tau_2\tau_3\tau_1\tau_2$ $\qquad$ $(1\,3) = \tau_1\tau_2\tau_1$

Figure 1.3: A permutation as a braid of strands, read from bottom to top and threaded from elementary layers. Disjoint layers, such as τ_1 and τ_3, are drawn at one level. The number of crossings equals $\operatorname{inv}(\sigma)$.

Definition 1.18 (Sign of a permutation). The *sign* of a permutation $\sigma \in S_n$ is

$$\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)} \in \{+1, -1\},$$

where $\operatorname{inv}(\sigma)$ is the number of inversions, that is, of crossings of strands.

Proposition 1.19. *Every transposition has sign -1, and $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$ for all $\sigma, \tau \in S_n$. Hence if σ is a product of k transpositions then $\operatorname{sgn}(\sigma) = (-1)^k$, and the parity of k does not depend on the decomposition.*

Proof ($\star$). The key step is this: adding one elementary layer flips the sign. Indeed, a layer τ_i on top swaps just two neighbouring strands, and every other pair keeps its order. So exactly one crossing appears or disappears in the braid:

$$\operatorname{inv}(\tau_i\sigma) = \operatorname{inv}(\sigma) \pm 1, \qquad \text{hence} \qquad \operatorname{sgn}(\tau_i\sigma) = -\operatorname{sgn}(\sigma).$$

Starting from $\sigma = e$, for which $\mathrm{inv}(e) = 0$, and repeating this k times: a braid of k elementary layers has sign $(-1)^k$.

Any transposition $(a\ b)$ with $a < b$ can be composed from elementary transpositions:

$$(a\ b) = \tau_a \cdots \tau_{b-2}\, \tau_{b-1}\, \tau_{b-2} \cdots \tau_a.$$

This composition swaps strand a with strand $a+1$, then with $a+2$, and so on all the way to strand b; the remaining elementary swaps pull strand b back to the starting place of a. There are $2(b-a)-1$ layers in all, an *odd* number, so $\mathrm{sgn}\big((a\ b)\big) = -1$.

Finally let σ be a product of k transpositions and τ a product of l (such decompositions exist by Proposition 1.17). Each transposition is an odd number of elementary layers, so $\mathrm{sgn}(\sigma) = (-1)^k$, $\mathrm{sgn}(\tau) = (-1)^l$ and $\mathrm{sgn}(\sigma\tau) = (-1)^{k+l} = \mathrm{sgn}(\sigma)\,\mathrm{sgn}(\tau)$. This also shows that the parity of k is independent of the decomposition: it is determined by $\mathrm{sgn}(\sigma)$ itself. ■

Remark 1.20. There is an equivalent algebraic definition of the sign of a permutation. Permutations can act on polynomials by relabelling the variable indices. Then $\mathrm{sgn}(\sigma)$ records how the polynomial $\Delta = \prod_{i<j}(x_i - x_j)$ changes. We take up this action on functions in Lecture 3.

Permutations of sign $+1$ are called *even*, those of sign -1 *odd*. A k-cycle is a product of $k-1$ transpositions (Proposition 1.17), hence has sign $(-1)^{k-1}$. The sign of a permutation will reappear throughout the course. Most immediately in Lecture 3, where sgn will be our first example of a nontrivial one-dimensional representation. Later, in Part II, it appears in the guise of the Levi-Civita symbol.

1.3.4 Comparing groups: isomorphism

The rotations through multiples of $2\pi/3$, the cube roots of unity, and the residues $\{0, 1, 2\}$ modulo 3 are "the same group" in three different guises. Their elements can be paired up so that the multiplication tables coincide. The following definition makes this precise.

Definition 1.21 (Isomorphism)**.** An *isomorphism* between groups G and H is a bijection $\varphi\colon G \to H$ that preserves multiplication:

$$\varphi(g_1 g_2) = \varphi(g_1)\,\varphi(g_2) \qquad \text{for all } g_1, g_2 \in G.$$

If an isomorphism exists, G and H are *isomorphic*, written $G \cong H$.

An isomorphism automatically matches the identity with the identity, and inverses with inverses. From $\varphi(e)\varphi(e) = \varphi(e)$ and cancellation in H we get $\varphi(e) = e_H$. Then $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$ gives $\varphi(g^{-1}) = \varphi(g)^{-1}$.

Example 1.22. The three guises of C_n in Section 1.3.1 are isomorphic. Write ρ_k for the rotation through $2\pi k/n$, and $\omega = e^{2\pi \mathrm{i}/n}$. The maps

$$k \longmapsto \rho_k \longmapsto \omega^k$$

are bijections $\mathbb{Z}_n \to \{\rho_k\} \to \{\omega^k\}$. The first converts addition modulo n into composition of rotations, because $\rho_a\rho_b = \rho_{a+b}$. The second converts composition of rotations into multiplication of complex numbers, because $\omega^a\omega^b = \omega^{a+b}$; the powers are read modulo n in both cases.

More generally, any two cyclic groups of the same order n are isomorphic via $g^k \mapsto h^k$. There is essentially *one* cyclic group of each order, which is why the notation C_n mentions only n.

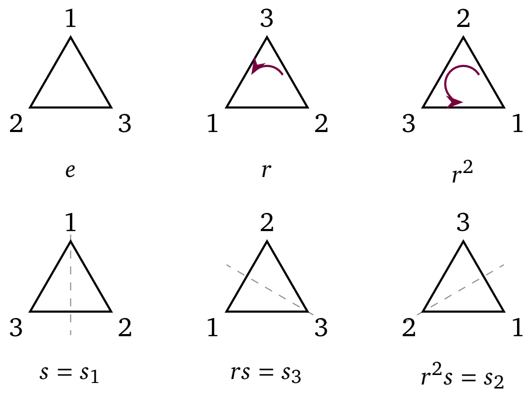


Figure 1.4: All six symmetry operations of the triangle. The number at a corner shows which original vertex ends up there. Rotations on top, reflections below with their axes dashed.

Example 1.23 ($D_3 \cong S_3$)**.** Every symmetry operation of the triangle permutes the vertices $\{1, 2, 3\}$. Sending each operation to the vertex permutation it induces gives a map $\varphi\colon D_3 \to S_3$. All six values can be read off Figure 1.4:

$$\begin{aligned} &e \mapsto e, && r \mapsto (1\,2\,3), && r^2 \mapsto (1\,3\,2), \\ &s \mapsto (2\,3), && rs \mapsto (1\,2), && r^2s \mapsto (1\,3). \end{aligned}$$

All six images are distinct and $|S_3| = 3! = 6$, so φ is a bijection. It remains to check that it preserves multiplication.

The vertex labels travel with the triangle, so performing two operations in succession multiplies the permutations in the same order (Figure 1.5). For instance $sr = r^2s$, so the table gives $\varphi(sr) = (1\,3)$. On the other hand $\varphi(s)\varphi(r) = (2\,3)(1\,2\,3) = (1\,3)$. Hence $\varphi(gh) = \varphi(g)\varphi(h)$. All the remaining cases are checked the same way. This shows that φ is an isomorphism of groups.

For $n \geq 4$ the same map embeds D_n into S_n (written $D_n \hookrightarrow S_n$), but is no longer surjective: $2n < n!$. Indeed not every permutation of the vertices of a square is realised by a rigid motion.

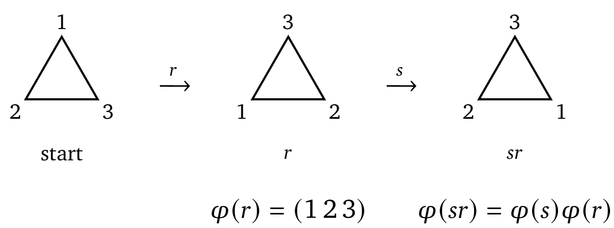


Figure 1.5: Why φ preserves multiplication. Performing r and then s gives the same result as applying sr at once. The labels travel with the triangle, so the permutations multiply in the same order.

Remark 1.24 (Invariants, and the classification problem). Isomorphic groups have all the same properties that can be expressed using the multiplication alone: the same order $|G|$, the same abelianness, the same orders of elements. Such properties are called *invariants*.

Invariants are the easy way to show that two groups are *not* isomorphic. On exercise sheet 1 you will check that the order-8 groups D_4 and C_8 are distinguished by their element orders — so they cannot be isomorphic. Classifying all finite groups up to isomorphism is one of the great problems of finite group theory. The reading at the end of this lecture surveys it briefly.

Reading

1.4 Groups beyond the finite

Part I of this course concerns finite groups, but we can define the main matrix groups of Part II already now. They show where the finite groups sit among all groups.

Definition 1.25 (Matrix groups). Let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$. The *general linear group* $\mathrm{GL}(n,\mathbb{F})$ is the set of invertible $n\times n$ matrices over $\mathbb{F}$ under matrix multiplication. Inside it sit, among others,

$$\begin{aligned} \mathrm{O}(n) &= \{A\in \mathrm{GL}(n,\mathbb{R}) : A^{\mathsf{T}}A = \mathbb{1}\}, &\qquad \mathrm{SO}(n) &= \{A\in \mathrm{O}(n) : \det A = 1\},\\ \mathrm{U}(n) &= \{A\in \mathrm{GL}(n,\mathbb{C}) : A^{\dagger}A = \mathbb{1}\}, &\qquad \mathrm{SU}(n) &= \{A\in \mathrm{U}(n) : \det A = 1\}, \end{aligned}$$

the *orthogonal*, *special orthogonal*, *unitary* and *special unitary* groups.

Example 1.26 ($\mathrm{O}(n)$ is a group). Matrix multiplication is associative and $\mathbb{1}^{\mathsf{T}}\mathbb{1} = \mathbb{1}$. So it suffices to check closure and inverses inside $\mathrm{GL}(n,\mathbb{R})$.

Closure:

$$A^{\mathsf{T}}A = B^{\mathsf{T}}B = \mathbb{1} \quad\Longrightarrow\quad (AB)^{\mathsf{T}}(AB) = B^{\mathsf{T}}A^{\mathsf{T}}AB = B^{\mathsf{T}}B = \mathbb{1}.$$

Inverses:

$$A^{\mathsf{T}}A = \mathbb{1} \quad\Longrightarrow\quad A^{-1} = A^{\mathsf{T}} \quad\Longrightarrow\quad (A^{-1})^{\mathsf{T}}A^{-1} = AA^{\mathsf{T}} = AA^{-1} = \mathbb{1}.$$

(For square matrices the relation $A^{\mathsf{T}}A = \mathbb{1}$ implies $AA^{\mathsf{T}} = \mathbb{1}$.)

The same manipulations with $\dagger$ in place of T show that $\mathrm{U}(n)$ is a group. For determinants $\det(AB) = \det A \det B$ and $\det A^{-1} = 1/\det A$, so $\mathrm{SO}(n)$ and $\mathrm{SU}(n)$ are groups too.

These matrix groups are *continuous*: their elements are described by continuous parameters. For instance every element of $\mathrm{SO}(2)$ is a rotation matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad R(\theta)R(\theta') = R(\theta+\theta'),$$

parametrised by an angle $\theta \in [0, 2\pi)$. The group $\mathrm{O}(2)$ consists of these rotations together with the reflections in all lines through the origin.

The finite groups we have met fit inside these groups. The rotations of the regular n-gon form a copy of C_n inside $\mathrm{SO}(2)$, and all its symmetry operations form a copy of D_n inside $\mathrm{O}(2)$. In the plane there are no other possibilities.

Remark 1.27 (Leonardo's theorem)**.** Every finite group of orthogonal transformations of the plane is cyclic or dihedral. This result is traditionally attributed to Leonardo da Vinci, who catalogued the possible symmetries of buildings.

For the statement to hold as written, the dihedral family must be extended downwards by two members, D_1 and D_2. The group D_1 consists of the identity and one reflection, so it has order 2. The group D_2 consists of two perpendicular reflections and the half-turn; it is the Klein four-group, which is the symmetry group of water and reappears in Lecture 5. Definition 1.9 does not name these groups, because no n-gon with $n < 3$ realises them.

The analogous classification of symmetries in three-dimensional space is taken up in Lecture 2. It determines the possible shapes of molecules and underlies crystallography.

1.5 Subgroups

Symmetry groups offer natural subsets of their own: the rotations inside D_n, the even permutations inside S_n. These are groups in their own right, sitting inside a larger one.

Definition 1.28 (Subgroup)**.** A subset $H \subseteq G$ of a group is a *subgroup*, written $H \leq G$, if H is itself a group under the inherited multiplication.

Proposition 1.29 (Subgroup test)**.** *A subset $H \subseteq G$ is a subgroup if and only if*

(i) $H \neq \emptyset$;

(ii) $h, k \in H$ implies $hk \in H$ (closure);

(iii) $h \in H$ implies $h^{-1} \in H$.

If G is finite, conditions (i) and (ii) suffice.

Proof. The necessity of all three conditions is clear. Note only that $e_H = e$: the identity of H satisfies $e_H e_H = e_H$, and cancellation in G gives $e_H = e$.

Conversely, suppose all three hold. Associativity is inherited from G. Pick any $h \in H$: closure and the inverse condition give $e = hh^{-1} \in H$. So H is a group.

Let G be finite, and assume only the first two conditions. Take $h \in H$. Closure puts all powers $h, h^2, h^3, \ldots$ in H. By Proposition 1.7 the element h has some finite order n, so $h^n = e \in H$ and $h^{n-1} = h^{-1} \in H$. That is the inverse condition.

Note that the last claim holds only for finite groups. In an infinite group closure alone is not enough: the set $\mathbb{N} \subset \mathbb{Z}$ is closed under addition, but is not a subgroup of $\mathbb{Z}$, since it has no inverses. ■

Example 1.30. (i) Every group has the *trivial* subgroup $\{e\}$ and the whole group G itself. Subgroups other than G are called *proper*.

(ii) The n rotations $\{e, r, \ldots, r^{n-1}\}$ in D_n form a subgroup isomorphic to C_n, since products and inverses of rotations are rotations.

(iii) For any $g \in G$ the set of all powers $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$ is a subgroup. It is called the *cyclic subgroup generated by* g. If g has finite order then $|\langle g \rangle| = \mathrm{ord}(g)$, since the distinct powers are exactly $e, g, \ldots, g^{\mathrm{ord}(g)-1}$.

(iv) The even permutations form a subgroup $A_n \leq S_n$, called the *alternating group*. Indeed sgn is multiplicative (Proposition 1.19), so products of even permutations are even, and it remains to apply the subgroup test for finite groups. For $n \geq 2$ exactly half of all permutations are even, so $|A_n| = n!/2$. To see this, fix a transposition τ: the map $\sigma \mapsto \tau\sigma$ is a bijection from even permutations to odd ones. The group A_4, of order 12, will appear again and again in this course; it is the rotation group of the regular tetrahedron.

1.6 Cosets and Lagrange's theorem

How large can a subgroup of a finite group be? The answer is strikingly rigid. The idea of the proof — slicing a group into translates of a subgroup — will be used constantly in everything that follows.

Definition 1.31 (Coset)**.** Let $H \leq G$ and $g \in G$. The *left coset* of H by g is the set

$$gH = \{\, gh : h \in H \,\} \subseteq G.$$

Right cosets Hg are defined analogously. In Lecture 2 the comparison of the two will lead to the notion of a *normal* (invariant) subgroup.

Example 1.32. In $G = D_3$ take $H = \langle s \rangle = \{e, s\}$. The left cosets are

$$eH = \{e, s\}, \qquad rH = \{r, rs\}, \qquad r^2H = \{r^2, r^2s\},$$

and no others: for instance $sH = \{s, e\} = eH$ and $(rs)H = \{rs, r\} = rH$. The three cosets are disjoint, each has two elements, and together they exhaust the six elements of D_3. So we have split D_3 into three cosets: $D_3 = eH \cup rH \cup r^2H$.

Lemma 1.33. *Let $H \leq G$. Then:*

(i) $gH = g'H$ *if and only if* $g^{-1}g' \in H$;

(ii) any two left cosets are either equal or disjoint, and the cosets partition G;

(iii) every coset has exactly $|H|$ elements (when H is finite).

Proof. (i) Suppose $g^{-1}g' = h_0 \in H$. Then $g'H = gh_0H = gH$, since $h_0H = H$. Indeed, the map $h \mapsto h_0h$ sends H into H by closure and is injective by cancellation, and it is onto H because $h \mapsto h_0^{-1}h$ is an inverse for it.

Conversely, suppose $gH = g'H$. Then $g' = g'e \in g'H = gH$, so $g' = gh$ for some $h \in H$, i.e. $g^{-1}g' = h \in H$.

(ii) Suppose $gH \cap g'H \neq \emptyset$, say $gh = g'h'$. Then $g^{-1}g' = h(h')^{-1} \in H$, so $gH = g'H$ by the first part. Since $g = ge \in gH$, every element lies in some coset. Hence the distinct cosets partition G.

(iii) The map $H \to gH$, $h \mapsto gh$, is surjective by definition and injective by cancellation. ∎

Theorem 1.34 (Lagrange)**.** *If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. More precisely, if $[G : H]$ denotes the number of distinct left cosets of H in G — the* index *of H — then*

$$|G| = [G : H] \cdot |H|\,.$$

Proof. By Lemma 1.33, G is partitioned into $[G : H]$ cosets, each of size exactly $|H|$. ∎

Corollary 1.35. *Let G be a finite group. Then:*

(i) $\operatorname{ord}(g)$ *divides $|G|$ for every $g \in G$;*

(ii) $g^{|G|} = e$ *for every $g \in G$;*

(iii) if $|G| = p$ is prime, then $G \cong C_p$; in particular G is cyclic and abelian.

Proof. (i) $\mathrm{ord}(g) = |\langle g\rangle|$, which divides $|G|$ by Lagrange. (ii) Write $|G| = \mathrm{ord}(g)\, m$; then $g^{|G|} = (g^{\mathrm{ord}(g)})^m = e$. (iii) Pick $g \neq e$. Then $|\langle g\rangle| > 1$ divides p, hence equals p. So $G = \langle g\rangle$ is cyclic of order p, and by Example 1.22 it is isomorphic to C_p. ■

For us, Lagrange's theorem is above all a constraint. It sharply narrows the choice of possible subgroups and element orders. Lecture 4 will add a constraint of the same kind on the dimensions of the *irreducible representations* (certain matrices that record the symmetry operations).

Example 1.36 (All subgroups of D_3)**.** A subgroup of D_3 can have order 1, 2, 3 or 6:

- (i) Order 1: only $\{e\}$.
- (ii) Order 2: the subgroup is generated by an element of order 2 (Corollary 1.35 with $p = 2$), that is, by a reflection. This gives $\langle s\rangle$, $\langle rs\rangle$, $\langle r^2 s\rangle$.
- (iii) Order 3: the subgroup is cyclic, generated by an element of order 3. The only such elements are r, r^2, and they generate the same subgroup $\langle r\rangle = \{e, r, r^2\}$.
- (iv) Order 6: D_3 itself.

So D_3 has exactly six subgroups. You will carry out the same census for D_4 on exercise sheet 1.

Remark 1.37. The converse of Lagrange's theorem is false: a divisor of $|G|$ need not be the order of any subgroup. The smallest counterexample is the alternating group A_4, of order 12, which has no subgroup of order 6. We state this without proof, since checking it needs more than we have learned in this lecture. Partial converses do exist. The most useful is Cauchy's theorem: a prime divisor of $|G|$ is always the order of some element. The case $p = 2$ will be on exercise sheet 1.

Reading

1.7 Outlook: the landscape of finite groups

How many finite groups are there? Counting makes sense only up to isomorphism, as Remark 1.24 explained. The numbers then begin as follows.

order n	1	2	3	4	5	6	7	8	9	10	11	12
# groups	1	1	1	2	1	2	1	5	2	2	1	5

The entries 1 at prime order are Corollary 1.35: for every prime p there is just one group of order p, namely C_p. At order 4 there are two groups: C_4 and the Klein four-group, water's own symmetry group. At order 6 there are again two, C_6 and $S_3 \cong D_3 \cong C_{3v}$. The second is the smallest nonabelian group, and this lecture has examined it from three directions.

The number of groups varies very irregularly, and jumps at powers of 2. There are 14 groups of order 16. Of all groups of order at most 2000, most have order exactly 1024.

A deeper organising principle is developed in Lecture 2. Finite groups are built from *simple* groups, that is, from groups admitting no nontrivial quotients. This is somewhat as integers are built from primes, and molecules from atoms.

The classification of the finite simple groups was completed in the late twentieth century, after tens of thousands of journal pages. It is among the largest collaborative achievements in mathematics. Its strangest entry is the Monster group, with about 8×10^{53} elements; it has unexpected connections to string theory.

None of this is needed in this course. Still, it is worth knowing what sort of territory we are taking our first steps in. In Lecture 2 we study how groups map to one another and how they decompose. From Lecture 3 the central theme becomes how groups *act on linear spaces*. That is precisely the form in which symmetry enters quantum mechanics.

Summary.

- *Group.* A set with an associative product, an identity, and inverses (Definition 1.1); abelian when the product commutes.
- *Standard families.* The cyclic groups, the dihedral group D_n of order $2n$ (symmetries of the n-gon, Proposition 1.10), and the symmetric group S_n with its multiplicative sign map (a *homomorphism*, in Lecture 2's language) (Definition 1.18).
- *Isomorphism.* A structure-preserving bijection; groups are studied up to isomorphism (Definition 1.21).
- *Subgroups and cosets.* The subgroup test (Proposition 1.29); cosets partition a group into blocks of equal size (Lemma 1.33).
- *Lagrange.* $|H|$ divides $|G|$ for $H \leq G$ (Theorem 1.34); hence element orders divide $|G|$ and groups of prime order are cyclic (Corollary 1.35).

Tutorial. Exercise Sheet 1.

References

Armstrong, *Groups and Symmetry* (groups, subgroups, Lagrange's theorem); Jones, *Groups, Representations and Physics*, Pt. 1; Tung, *Group Theory in Physics*, Ch. 1.